
Algorithm 3 Binary Search with Backtracking (α, δ, n, m)

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1: Initialize:  $b_{max}$  and  $t_{max} = \lfloor \frac{n}{b_{max}} \rfloor$ , Thershold  $\tau$ 
2: Divide  $n$  users into  $t_{max}$  set of users (say  $Z_t$ ), each of size  $b_{max}$  and set  $L = -1$  and  $R = 1$ 
3: Initialize a list for end points  $LR\_list = [ [L, R] ]$ 
4: for  $t \leftarrow 1$  to  $t_{max}$  do
5:   backtrack = False
6:   if  $t > 1$  then
7:     for  $u \in Z_t$  do

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$$C_t^{(u)} = 1_{\{\bar{X}^{(u)} \in [L_t, R_t]\}} \quad \tilde{C}_t^{(u)} = \begin{cases} C_t^{(u)}, & \text{w.p. } \frac{e^{\alpha/2}}{1+e^{\alpha/2}} \\ 1 - C_t^{(u)}, & \text{otherwise} \end{cases}$$

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8:   end for
9:    $\mathcal{C}_t = \sum_{u=1}^{b_{max}} \tilde{C}_t^{(u)}$ 
10:  if  $\mathcal{C} < \tau$  then
11:     $LR\_list.pop()$ 
12:     $L, R = LR\_list[-1]$ 
13:    backtrack = True
14:  end if
15: end if
16: if not backtrack then
17:    $L_{prev}, R_{prev} = L, R$ 
18:   Divide  $[L, R]$  into three equal-length intervals. Let  $\gamma_t = \frac{R-L}{3}$ .

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$$I_1 = [L, L + \gamma_t], \quad I_2 = [L + \gamma_t, R - \gamma_t], \quad I_3 = [R - \gamma_t, R]$$

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19:  for  $u \in Z_t$  do

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$$V_{I_1,t}^{(u)} = 1_{\{\bar{X}^{(u)} \in I_1\}} \quad \tilde{V}_{I_1,t}^{(u)} = \begin{cases} V_{I_1,t}^{(u)}, & \text{w.p. } \frac{e^{\alpha/2}}{1+e^{\alpha/2}} \\ 1 - V_{I_1,t}^{(u)}, & \text{otherwise} \end{cases}$$

and

$$V_{I_3,t}^{(u)} = 1_{\{\bar{X}^{(u)} \in I_3\}} \quad \tilde{V}_{I_3,t}^{(u)} = \begin{cases} V_{I_3,t}^{(u)}, & \text{w.p. } \frac{e^{\alpha/2}}{1+e^{\alpha/2}} \\ 1 - V_{I_3,t}^{(u)}, & \text{otherwise} \end{cases}$$

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20:  end for
21:   $\mathcal{V}_{I_1,t} = \sum_{u=1}^{b_{max}} \tilde{V}_{I_1,t}^{(u)}$  and  $\mathcal{V}_{I_3,t} = \sum_{u=1}^{b_{max}} \tilde{V}_{I_3,t}^{(u)}$ 
22:  if  $V_{I_1,t} \geq V_{I_3,t}$  then
23:    Remove the interval  $I_3$ 
24:     $R = R - \gamma_t$ 
25:  else
26:    Remove the interval  $I_1$ 
27:     $L = L + \gamma_t$ 
28:  end if
29:   $LR\_list.append([L, R])$ 
30: end if
31: end for
32: return  $[L, R]$ 

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5 DAME with Binary Search with Backtracking

Algorithm 4 DAME with Binary Search with Backtracking

- 1: **Input:** number of users (n), privacy parameter (α), number of samples per user (m)
 - 2: **Initialize:** $scale$ (for laplacian noise), δ (failure probability of binary search) and δ' (length of expansion)
 - 3: **Localization Phase:**
 - 4: $[L, R] = Binary_Search_with_Backtracking(\alpha, \delta, n/2, m)$
 - 5: Define $\tilde{L} = L - \delta'$, $\tilde{R} = R + \delta'$ and $\mathcal{I} = [\tilde{L}, \tilde{R}]$
 - 6: **Estimation phase:**
 - 7: **for** $u \in [\frac{n}{2} + 1, n]$ **do**
 - 8: Sample l_u from standard Laplace distribution.
 - 9: Share $\hat{\theta}^{(u)} = \Pi_{\mathcal{I}}(\bar{X}^{(u)}) + (scale) l_u$.
 - 10: **end for**
 - 11: Return $\hat{\theta} = \frac{2}{n} \sum_{u=\frac{n}{2}+1}^n \hat{\theta}^{(u)}$.
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